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Running Tune experiments with AxSearch

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In this tutorial we introduce Ax, while running a simple Ray Tune experiment. Tune's Search Algorithms integrate with Ax and, as a result, allow you to seamlessly scale up a Ax optimization process - without sacrificing performance.

Ax is a platform for optimizing any kind of experiment, including machine learning experiments, A/B tests, and simulations. Ax can optimize discrete configurations (e.g., variants of an A/B test) using multi-armed bandit optimization, and continuous/ordered configurations (e.g. float/int parameters) using Bayesian optimization. Results of A/B tests and simulations with reinforcement learning agents often exhibit high amounts of noise. Ax supports state-of-the-art algorithms which work better than traditional Bayesian optimization in high-noise settings. Ax also supports multi-objective and constrained optimization which are common to real-world problems (e.g. improving load time without increasing data use). Ax belongs to the domain of "derivative-free" and "black-box" optimization.

In this example we minimize a simple objective to briefly demonstrate the usage of AxSearch with Ray Tune via `AxSearch`. It's useful to keep in mind that despite the emphasis on machine learning experiments, Ray Tune optimizes any implicit or explicit objective. Here we assume `ax-platform==0.2.4` library is installed with the python version ≥ 3.7 . To learn more, please refer to the [Ax website](#).

Click below to see all the imports we need for this example.

[▶ Show code cell source](#)

Let's start by defining a classic benchmark for global optimization. The form here is explicit for demonstration, yet it is typically a black-box. We artificially sleep for a bit (`0.02` seconds) to simulate a long-running ML experiment. This setup assumes that we're running multiple `step`s of an experiment and try to tune 6-dimensions of the `x` hyperparameter.

```
def landscape(x):
    """
    Hartmann 6D function containing 6 local minima.
    It is a classic benchmark for developing global optimization algorithms.
    """
    alpha = np.array([1.0, 1.2, 3.0, 3.2])
    A = np.array(
        [
            [10, 3, 17, 3.5, 1.7, 8],
            [0.05, 10, 17, 0.1, 8, 14],
            [3, 3.5, 1.7, 10, 17, 8],
            [17, 8, 0.05, 10, 0.1, 14],
        ]
    )
    P = 10 ** (-4) * np.array(
        [
            [1312, 1696, 5569, 124, 8283, 5886],
            [2329, 4135, 8307, 3736, 1004, 9991],
            [2348, 1451, 3522, 2883, 3047, 6650],
            [4047, 8828, 8732, 5743, 1091, 381],
        ]
    )
    y = 0.0
    for j, alpha_j in enumerate(alpha):
        t = 0
        for k in range(6):
            t += A[j, k] * ((x[k] - P[j, k]) ** 2)
        y -= alpha_j * np.exp(-t)
    return y
```

Next, our `objective` function takes a Tune `config`, evaluates the `landscape` of our experiment in a training loop, and uses `tune.report` to report the `landscape` back to Tune.

```
def objective(config):
    for i in range(config["iterations"]):
        x = np.array([config.get("x{}".format(i + 1)) for i in range(6)])
        tune.report(
            {"timesteps_total": i, "landscape": landscape(x), "l2norm": np.sqrt((x ** 2).sum())}
        )
        time.sleep(0.02)
```

Next we define a search space. The critical assumption is that the optimal hyperparameters live within this space. Yet, if the space is very large, then those hyperparameters may be difficult to find in a short amount of time.

```
search_space = {
    "iterations": 100,
    "x1": tune.uniform(0.0, 1.0),
    "x2": tune.uniform(0.0, 1.0),
    "x3": tune.uniform(0.0, 1.0),
    "x4": tune.uniform(0.0, 1.0),
    "x5": tune.uniform(0.0, 1.0),
    "x6": tune.uniform(0.0, 1.0)
}
```

Now we define the search algorithm from `AxSearch`. If you want to constrain your parameters or even the space of outcomes, that can be easily done by passing the arguments as below.

```
algo = AxSearch(
    parameter_constraints=["x1 + x2 <= 2.0"],
    outcome_constraints=["l2norm <= 1.25"],
)
```

We also use `ConcurrencyLimiter` to constrain to 4 concurrent trials.

```
algo = tune.search.ConcurrencyLimiter(algo, max_concurrent=4)
```

The number of samples is the number of hyperparameter combinations that will be tried out. This Tune run is set to 1000 samples. You can decrease this if it takes too long on your machine, or you can set a time limit easily through stop argument in the RunConfig() as we will show here.

```
num_samples = 100
stop_timesteps = 200
```

Finally, we run the experiment to find the global minimum of the provided landscape (which contains 5 false minima). The argument to metric, "landscape", is provided via the objective function's tune.report. The experiment "min" imizes the "mean_loss" of the landscape by searching within search_space via algo, num_samples times or when "timesteps_total": stop_timesteps. This previous sentence is fully characterizes the search problem we aim to solve. With this in mind, notice how efficient it is to execute tuner.fit().

```
tuner = tune.Tuner(
    objective,
    tune_config=tune.TuneConfig(
        metric="landscape",
        mode="min",
        search_alg=algo,
        num_samples=num_samples,
    ),
    run_config=tune.RunConfig(
        name="ax",
        stop={"timesteps_total": stop_timesteps}
    ),
    param_space=search_space,
)
results = tuner.fit()
```

```
[INFO 07-22 15:04:18] ax.service.ax_client: Starting optimization with verbose logging. To disable logging, set the `verbose`
[INFO 07-22 15:04:18] ax.service.utils.instantiation: Created search space: SearchSpace(parameters=[FixedParameter(name='iter
[INFO 07-22 15:04:18] ax.modelbridge.dispatch_utils: Using Bayesian optimization since there are more ordered parameters than
[INFO 07-22 15:04:18] ax.modelbridge.dispatch_utils: Using Bayesian Optimization generation strategy:GenerationStrategy(name:
Detected sequential enforcement. Be sure to use a ConcurrencyLimiter.
```

== Status ==
Current time: 2022-07-22 15:04:35 (running for 00:00:16.56)
Memory usage on this node: 9.9/16.0 GiB
Using FIFO scheduling algorithm.
Resources requested: 0/16 CPUs, 0/0 GPUs, 0.0/5.13 GiB heap, 0.0/2.0 GiB objects
Current best trial: 34b7abda with landscape=-1.6624439263544026 and parameters={'iterations': 100, 'x1': 0.26526361983269453, 'x2': 0.9248840995132923, 'x3': 0.15171580761671066, 'x4': 0.43602637108415365, 'x5': 0.8573104059323668, 'x6': 0.08981018699705601}
Result logdir: /Users/kai/ray_results/ax
Number of trials: 10/10 (10 TERMINATED)

Trial name	status	loc	iterations	x1	x2	x3	x4	x5	
objective_2dfbe86a	TERMINATED	127.0.0.1:44721	100	0.0558336	0.0896192	0.958956	0.234474	0.174516	0.970
objective_2fa776c0	TERMINATED	127.0.0.1:44726	100	0.744772	0.754537	0.0950125	0.273877	0.0966829	0.3689
objective_2fabaa1a	TERMINATED	127.0.0.1:44727	100	0.405704	0.374626	0.935628	0.222185	0.787212	0.008124
objective_2faee7c0	TERMINATED	127.0.0.1:44728	100	0.664728	0.207519	0.359514	0.704578	0.755882	0.8124
objective_313d3d3a	TERMINATED	127.0.0.1:44747	100	0.0418746	0.992783	0.906027	0.594429	0.825393	0.6463
objective_32c9acd8	TERMINATED	127.0.0.1:44726	100	0.126064	0.703408	0.344681	0.337363	0.401396	0.6792

objective_32cf8ca2	TERMINATED	127.0.0.1:44756	100	0.0910936	0.304138	0.869848	0.405435	0.567922	0.2286
objective_32d8dd20	TERMINATED	127.0.0.1:44758	100	0.603178	0.409057	0.729056	0.0825984	0.572948	0.5083
objective_34adf04a	TERMINATED	127.0.0.1:44768	100	0.454189	0.271772	0.530871	0.991841	0.691843	0.4723
objective_34b7abda	TERMINATED	127.0.0.1:44771	100	0.265264	0.924884	0.151716	0.436026	0.85731	0.08981

```
[INFO 07-22 15:04:19] ax.service.ax_client: Generated new trial 0 with parameters {'x1': 0.055834, 'x2': 0.089619, 'x3': 0.951
[INFO 07-22 15:04:22] ax.service.ax_client: Generated new trial 1 with parameters {'x1': 0.744772, 'x2': 0.754537, 'x3': 0.091
[INFO 07-22 15:04:22] ax.service.ax_client: Generated new trial 2 with parameters {'x1': 0.405704, 'x2': 0.374626, 'x3': 0.931
[INFO 07-22 15:04:22] ax.service.ax_client: Generated new trial 3 with parameters {'x1': 0.664728, 'x2': 0.207519, 'x3': 0.351
```

```
Result for objective_2dfbe86a:
date: 2022-07-22_15-04-22
done: false
experiment_id: 4ef8a12ac94a4f4fa483ec18e347967f
hostname: Kais-MacBook-Pro.local
iterations_since_restore: 1
l2norm: 1.3991721132671366
landscape: -0.8052333562869153
node_ip: 127.0.0.1
pid: 44721
time_since_restore: 0.00022912025451660156
time_this_iter_s: 0.00022912025451660156
time_total_s: 0.00022912025451660156
timestamp: 1658498662
timesteps_since_restore: 0
timesteps_total: 0
training_iteration: 1
trial_id: 2dfbe86a
warmup_time: 0.0035619735717773438
```

```
[INFO 07-22 15:04:24] ax.service.ax_client: Completed trial 0 with data: {'landscape': (-0.805233, None), 'l2norm': (1.399172
[INFO 07-22 15:04:24] ax.service.ax_client: Generated new trial 4 with parameters {'x1': 0.041875, 'x2': 0.992783, 'x3': 0.901
```

```
Result for objective_2faee7c0:
date: 2022-07-22_15-04-24
done: false
experiment_id: 3699644e85ac439cb7c1a36ed0976307
hostname: Kais-MacBook-Pro.local
iterations_since_restore: 1
l2norm: 1.530347488145437
landscape: -0.011983676977099367
node_ip: 127.0.0.1
pid: 44728
time_since_restore: 0.00022292137145996094
time_this_iter_s: 0.00022292137145996094
time_total_s: 0.00022292137145996094
timestamp: 1658498664
timesteps_since_restore: 0
timesteps_total: 0
training_iteration: 1
trial_id: 2faee7c0
warmup_time: 0.0027179718017578125
```

```
Result for objective_2fa776c0:
date: 2022-07-22_15-04-24
done: false
experiment_id: c555bfed13ac43e5b8c8e9f6d4b9b2f7
hostname: Kais-MacBook-Pro.local
iterations_since_restore: 1
l2norm: 1.1634068454629019
landscape: -0.11285961764770336
node_ip: 127.0.0.1
pid: 44726
```

```

time_since_restore: 0.000225067138671875
time_this_iter_s: 0.000225067138671875
time_total_s: 0.000225067138671875
timestamp: 1658498664
timesteps_since_restore: 0
timesteps_total: 0
training_iteration: 1
trial_id: 2fa776c0
warmup_time: 0.0026290416717529297

```

Result for objective_2dfbe86a:

```

date: 2022-07-22_15-04-24
done: true
experiment_id: 4ef8a12ac94a4f4fa483ec18e347967f
experiment_tag: 1_iterations=100,x1=0.0558,x2=0.0896,x3=0.9590,x4=0.2345,x5=0.1745,x6=0.9703
hostname: Kais-MacBook-Pro.local
iterations_since_restore: 100
l2norm: 1.3991721132671366
landscape: -0.8052333562869153
node_ip: 127.0.0.1
pid: 44721
time_since_restore: 2.573719024658203
time_this_iter_s: 0.00251619815826416
time_total_s: 2.573719024658203
timestamp: 1658498664
timesteps_since_restore: 0
timesteps_total: 99
training_iteration: 100
trial_id: 2dfbe86a
warmup_time: 0.0035619735717773438

```

Result for objective_2fabaa1a:

```

date: 2022-07-22_15-04-24
done: false
experiment_id: eb9287e4fe5f44c7868dc943e2642312
hostname: Kais-MacBook-Pro.local
iterations_since_restore: 1
l2norm: 1.3599537840291782
landscape: -0.11348012497414121
node_ip: 127.0.0.1
pid: 44727
time_since_restore: 0.00022077560424804688
time_this_iter_s: 0.00022077560424804688
time_total_s: 0.00022077560424804688
timestamp: 1658498664
timesteps_since_restore: 0
timesteps_total: 0
training_iteration: 1
trial_id: 2fabaa1a
warmup_time: 0.0025510787963867188

```

```

[INFO 07-22 15:04:27] ax.service.ax_client: Completed trial 3 with data: {'landscape': (-0.011984, None), 'l2norm': (1.530347, None), 'x1': 0.091094, 'x2': 0.304138, 'x3': 0.861094, 'x4': 0.2345, 'x5': 0.1745, 'x6': 0.9703}
[INFO 07-22 15:04:27] ax.service.ax_client: Generated new trial 5 with parameters {'x1': 0.126064, 'x2': 0.703408, 'x3': 0.3408, 'x4': 0.2345, 'x5': 0.1745, 'x6': 0.9703}
[INFO 07-22 15:04:27] ax.service.ax_client: Completed trial 1 with data: {'landscape': (-0.11286, None), 'l2norm': (1.163407, None), 'x1': 0.091094, 'x2': 0.304138, 'x3': 0.861094, 'x4': 0.2345, 'x5': 0.1745, 'x6': 0.9703}
[INFO 07-22 15:04:27] ax.service.ax_client: Generated new trial 6 with parameters {'x1': 0.091094, 'x2': 0.304138, 'x3': 0.861094, 'x4': 0.2345, 'x5': 0.1745, 'x6': 0.9703}
[INFO 07-22 15:04:27] ax.service.ax_client: Completed trial 2 with data: {'landscape': (-0.11348, None), 'l2norm': (1.359954, None), 'x1': 0.091094, 'x2': 0.304138, 'x3': 0.861094, 'x4': 0.2345, 'x5': 0.1745, 'x6': 0.9703}
[INFO 07-22 15:04:27] ax.service.ax_client: Generated new trial 7 with parameters {'x1': 0.603178, 'x2': 0.409057, 'x3': 0.721094, 'x4': 0.2345, 'x5': 0.1745, 'x6': 0.9703}

```

Result for objective_313d3d3a:

```

date: 2022-07-22_15-04-27
done: false
experiment_id: fa7afd557e154f8bebe4f54d8eedb3573
hostname: Kais-MacBook-Pro.local
iterations_since_restore: 1
l2norm: 1.805729990121368
landscape: -0.006779757704679272
node_ip: 127.0.0.1
pid: 44747
time_since_restore: 0.00021076202392578125
time_this_iter_s: 0.00021076202392578125
time_total_s: 0.00021076202392578125
timestamp: 1658498667

```

```

timestamp: 1658498667
timesteps_since_restore: 0
timesteps_total: 0
training_iteration: 1
trial_id: 313d3d3a
warmup_time: 0.0029790401458740234

```

Result for objective_2faee7c0:

```

date: 2022-07-22_15-04-27
done: true
experiment_id: 3699644e85ac439cb7c1a36ed0976307
experiment_tag: 4_iterations=100,x1=0.6647,x2=0.2075,x3=0.3595,x4=0.7046,x5=0.7559,x6=0.8124
hostname: Kais-MacBook-Pro.local
iterations_since_restore: 100
l2norm: 1.530347488145437
landscape: -0.011983676977099367
node_ip: 127.0.0.1
pid: 44728
time_since_restore: 2.6206929683685303
time_this_iter_s: 0.027359962463378906
time_total_s: 2.6206929683685303
timestamp: 1658498667
timesteps_since_restore: 0
timesteps_total: 99
training_iteration: 100
trial_id: 2faee7c0
warmup_time: 0.0027179718017578125

```

Result for objective_2fa776c0:

```

date: 2022-07-22_15-04-27
done: true
experiment_id: c555bfed13ac43e5b8c8e9f6d4b9b2f7
experiment_tag: 2_iterations=100,x1=0.7448,x2=0.7545,x3=0.0950,x4=0.2739,x5=0.0967,x6=0.3689
hostname: Kais-MacBook-Pro.local
iterations_since_restore: 100
l2norm: 1.1634068454629019
landscape: -0.11285961764770336
node_ip: 127.0.0.1
pid: 44726
time_since_restore: 2.6361019611358643
time_this_iter_s: 0.0264589786529541
time_total_s: 2.6361019611358643
timestamp: 1658498667
timesteps_since_restore: 0
timesteps_total: 99
training_iteration: 100
trial_id: 2fa776c0
warmup_time: 0.0026290416717529297

```

Result for objective_32c9acd8:

```

date: 2022-07-22_15-04-27
done: false
experiment_id: c555bfed13ac43e5b8c8e9f6d4b9b2f7
hostname: Kais-MacBook-Pro.local
iterations_since_restore: 1
l2norm: 1.1686440476629836
landscape: -0.9046216637367911
node_ip: 127.0.0.1
pid: 44726
time_since_restore: 0.00020194053649902344
time_this_iter_s: 0.00020194053649902344
time_total_s: 0.00020194053649902344
timestamp: 1658498667

timesteps_since_restore: 0
timesteps_total: 0
training_iteration: 1
trial_id: 32c9acd8
warmup_time: 0.0026290416717529297

```

Result for objective_2fabaa1a:

```

date: 2022-07-22_15-04-27
done: true
experiment_id: eb9287e4fe5f44c7868dc943e2642312
experiment_tag: 3_iterations=100,x1=0.4057,x2=0.3746,x3=0.9356,x4=0.2222,x5=0.7872,x6=0.0081
hostname: Kais-MacBook-Pro.local
iterations_since_restore: 100
l2norm: 1.3599537840291782
landscape: -0.112490112407414121

```

```

landscape: -0.11340012497414121
node_ip: 127.0.0.1
pid: 44727
time_since_restore: 2.623929977416992
time_this_iter_s: 0.032716989517211914
time_total_s: 2.623929977416992
timestamp: 1658498667
timesteps_since_restore: 0
timesteps_total: 99
training_iteration: 100
trial_id: 2fabaa1a
warmup_time: 0.0025510787963867188

```

Result for objective_32d8dd20:

```

date: 2022-07-22_15-04-30
done: false
experiment_id: 171527593b0f4cbf941c0a03faaf0953
hostname: Kais-MacBook-Pro.local
iterations_since_restore: 1
l2norm: 1.2869105702896437
landscape: -0.24722262157458608
node_ip: 127.0.0.1
pid: 44758
time_since_restore: 0.00021886825561523438
time_this_iter_s: 0.00021886825561523438
time_total_s: 0.00021886825561523438
timestamp: 1658498670
timesteps_since_restore: 0
timesteps_total: 0
training_iteration: 1
trial_id: 32d8dd20
warmup_time: 0.002732992172241211

```

Result for objective_32cf8ca2:

```

date: 2022-07-22_15-04-29
done: false
experiment_id: 37610500f6df493aae4e7e46bb21bf09
hostname: Kais-MacBook-Pro.local
iterations_since_restore: 1
l2norm: 1.1817810425508524
landscape: -0.14653248187442922
node_ip: 127.0.0.1
pid: 44756
time_since_restore: 0.00025081634521484375
time_this_iter_s: 0.00025081634521484375
time_total_s: 0.00025081634521484375
timestamp: 1658498669
timesteps_since_restore: 0
timesteps_total: 0
training_iteration: 1
trial_id: 32cf8ca2
warmup_time: 0.0032138824462890625

```

```

[INFO 07-22 15:04:30] ax.service.ax_client: Completed trial 4 with data: {'landscape': (-0.00678, None), 'l2norm': (1.80573, 1
[INFO 07-22 15:04:30] ax.service.ax_client: Generated new trial 8 with parameters {'x1': 0.454189, 'x2': 0.271772, 'x3': 0.53
[INFO 07-22 15:04:30] ax.service.ax_client: Completed trial 5 with data: {'landscape': (-0.904622, None), 'l2norm': (1.168644
[INFO 07-22 15:04:30] ax.service.ax_client: Generated new trial 9 with parameters {'x1': 0.265264, 'x2': 0.924884, 'x3': 0.15

```

```

Result for objective_313d3d3a:
date: 2022-07-22_15-04-30
done: true
experiment_id: fa7afd557e154fbeb4f54d8eedb3573
experiment_tag: 5_iterations=100,x1=0.0419,x2=0.9928,x3=0.9060,x4=0.5944,x5=0.8254,x6=0.6464
hostname: Kais-MacBook-Pro.local
iterations_since_restore: 100
l2norm: 1.805729990121368
landscape: -0.006779757704679272
node_ip: 127.0.0.1
pid: 44747
time_since_restore: 3.1623308658599854
time_this_iter_s: 0.02911996841430664
time_total_s: 3.1623308658599854
timestamp: 1658498670
timesteps_since_restore: 0
timesteps_total: 99
training_iteration: 100
trial_id: 313d3d3a
warmup_time: 0.0029790401458740234

```

```

Result for objective_32c9acd8:
date: 2022-07-22_15-04-30
done: true
experiment_id: c555bfed13ac43e5b8c8e9f6d4b9b2f7
experiment_tag: 6_iterations=100,x1=0.1261,x2=0.7034,x3=0.3447,x4=0.3374,x5=0.4014,x6=0.6792
hostname: Kais-MacBook-Pro.local
iterations_since_restore: 100
l2norm: 1.1686440476629836
landscape: -0.9046216637367911
node_ip: 127.0.0.1
pid: 44726
time_since_restore: 3.1211891174316406
time_this_iter_s: 0.02954697608947754
time_total_s: 3.1211891174316406
timestamp: 1658498670
timesteps_since_restore: 0
timesteps_total: 99
training_iteration: 100
trial_id: 32c9acd8
warmup_time: 0.0026290416717529297

```

```

[INFO 07-22 15:04:32] ax.service.ax_client: Completed trial 7 with data: {'landscape': (-0.247223, None), 'l2norm': (1.286911, None), 'time_total_s': 3.1623308658599854, 'time_this_iter_s': 0.02911996841430664, 'timesteps_total': 99, 'timesteps_since_restore': 0, 'training_iteration': 100, 'trial_id': '313d3d3a', 'warmup_time': 0.0029790401458740234}
[INFO 07-22 15:04:32] ax.service.ax_client: Completed trial 6 with data: {'landscape': (-0.146532, None), 'l2norm': (1.181781, None), 'time_total_s': 3.1211891174316406, 'time_this_iter_s': 0.02954697608947754, 'timesteps_total': 99, 'timesteps_since_restore': 0, 'training_iteration': 100, 'trial_id': '32c9acd8', 'warmup_time': 0.0026290416717529297}

```

```

Result for objective_32d8dd20:
date: 2022-07-22_15-04-32
done: true
experiment_id: 171527593b0f4cbf941c0a03faaf0953
experiment_tag: 8_iterations=100,x1=0.6032,x2=0.4091,x3=0.7291,x4=0.0826,x5=0.5729,x6=0.5083
hostname: Kais-MacBook-Pro.local
iterations_since_restore: 100
l2norm: 1.2869105702896437
landscape: -0.24722262157458608
node_ip: 127.0.0.1
pid: 44758
time_since_restore: 2.6415798664093018
time_this_iter_s: 0.026781082153320312
time_total_s: 2.6415798664093018
timestamp: 1658498672
timesteps_since_restore: 0
timesteps_total: 99
training_iteration: 100
trial_id: 32d8dd20
warmup_time: 0.002732992172241211

```

```

Result for objective_32cf8ca2:
date: 2022-07-22_15-04-32
done: true
experiment_id: 37610500f6df493aae4e7e46bb21bf09
experiment_tag: 7_iterations=100,x1=0.0911,x2=0.3041,x3=0.8698,x4=0.4054,x5=0.5679,x6=0.2286
hostname: Kais-MacBook-Pro.local
iterations_since_restore: 100
l2norm: 1.2869105702896437
landscape: -0.24722262157458608
node_ip: 127.0.0.1
pid: 44758
time_since_restore: 2.6415798664093018
time_this_iter_s: 0.026781082153320312
time_total_s: 2.6415798664093018
timestamp: 1658498672
timesteps_since_restore: 0
timesteps_total: 99
training_iteration: 100
trial_id: 32cf8ca2
warmup_time: 0.002732992172241211

```



```

hostname: Kais-MacBook-Pro.local
iterations_since_restore: 100
l2norm: 1.1817810425508524
landscape: -0.14653248187442922
node_ip: 127.0.0.1
pid: 44756
time_since_restore: 2.707913875579834
time_this_iter_s: 0.027456998825073242
time_total_s: 2.707913875579834
timestamp: 1658498672
timesteps_since_restore: 0
timesteps_total: 99
training_iteration: 100
trial_id: 32cf8ca2
warmup_time: 0.0032138824462890625

```

Result for objective_34adf04a:

```

date: 2022-07-22_15-04-33
done: false
experiment_id: 4f65c5b68f5c49d98fda388e37c83deb
hostname: Kais-MacBook-Pro.local
iterations_since_restore: 1
l2norm: 1.4991655675380078
landscape: -0.01329150870283869
node_ip: 127.0.0.1
pid: 44768
time_since_restore: 0.00021600723266601562
time_this_iter_s: 0.00021600723266601562
time_total_s: 0.00021600723266601562
timestamp: 1658498673
timesteps_since_restore: 0
timesteps_total: 0
training_iteration: 1
trial_id: 34adf04a
warmup_time: 0.0027239322662353516

```

Result for objective_34b7abda:

```

date: 2022-07-22_15-04-33
done: false
experiment_id: f135a2c40f5644ba9d2ae096a9dd10e0
hostname: Kais-MacBook-Pro.local
iterations_since_restore: 1
l2norm: 1.3718451333547932
landscape: -1.6624439263544026
node_ip: 127.0.0.1
pid: 44771
time_since_restore: 0.0002338886260986328
time_this_iter_s: 0.0002338886260986328
time_total_s: 0.0002338886260986328
timestamp: 1658498673

timesteps_since_restore: 0
timesteps_total: 0
training_iteration: 1
trial_id: 34b7abda
warmup_time: 0.002721071243286133

```

```

[INFO 07-22 15:04:35] ax.service.ax_client: Completed trial 8 with data: {'landscape': (-0.013292, None), 'l2norm': (1.499166
[INFO 07-22 15:04:35] ax.service.ax_client: Completed trial 9 with data: {'landscape': (-1.662444, None), 'l2norm': (1.371845

```

```

Result for objective_34adf04a:
date: 2022-07-22_15-04-35
done: true
experiment_id: 4f65c5b68f5c49d98fda388e37c83deb
experiment_tag: 9_iterations=100,x1=0.4542,x2=0.2718,x3=0.5309,x4=0.9918,x5=0.6918,x6=0.4724
hostname: Kais-MacBook-Pro.local
iterations_since_restore: 100
l2norm: 1.4991655675380078
landscape: -0.01329150870283869
node_ip: 127.0.0.1
pid: 44768
time_since_restore: 2.7032668590545654
time_this_iter_s: 0.029300928115844727
time_total_s: 2.7032668590545654
timestamp: 1658498675
timesteps_since_restore: 0
timesteps_total: 99
training_iteration: 100
trial_id: 34adf04a
warmup_time: 0.0027239322662353516

```

```

Result for objective_34b7abda:
date: 2022-07-22_15-04-35
done: true
experiment_id: f135a2c40f5644ba9d2ae096a9dd10e0
experiment_tag: 10_iterations=100,x1=0.2653,x2=0.9249,x3=0.1517,x4=0.4360,x5=0.8573,x6=0.0898
hostname: Kais-MacBook-Pro.local
iterations_since_restore: 100
l2norm: 1.3718451333547932
landscape: -1.6624439263544026
node_ip: 127.0.0.1
pid: 44771
time_since_restore: 2.6852078437805176
time_this_iter_s: 0.029579877853393555
time_total_s: 2.6852078437805176
timestamp: 1658498675
timesteps_since_restore: 0
timesteps_total: 99
training_iteration: 100
trial_id: 34b7abda
warmup_time: 0.002721071243286133

```

And now we have the hyperparameters found to minimize the mean loss.

```
print("Best hyperparameters found were: ", results.get_best_result().config)
```

```
Best hyperparameters found were: {'iterations': 100, 'x1': 0.26526361983269453, 'x2': 0.9248840995132923, 'x3': 0.1517158076}
```

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Yes

No

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